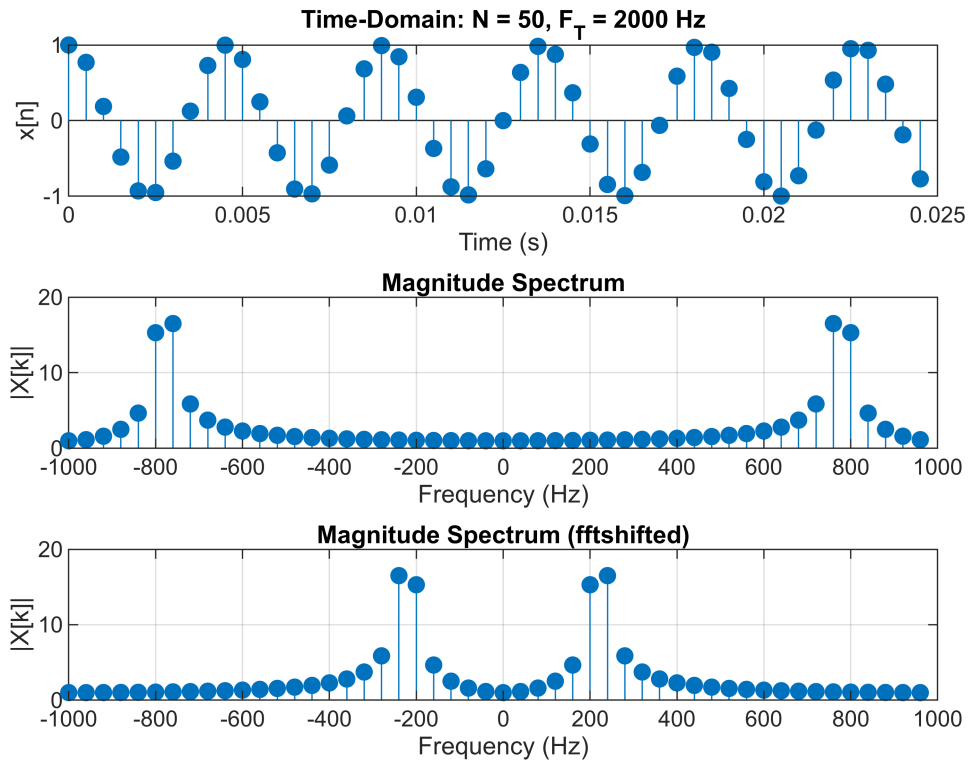
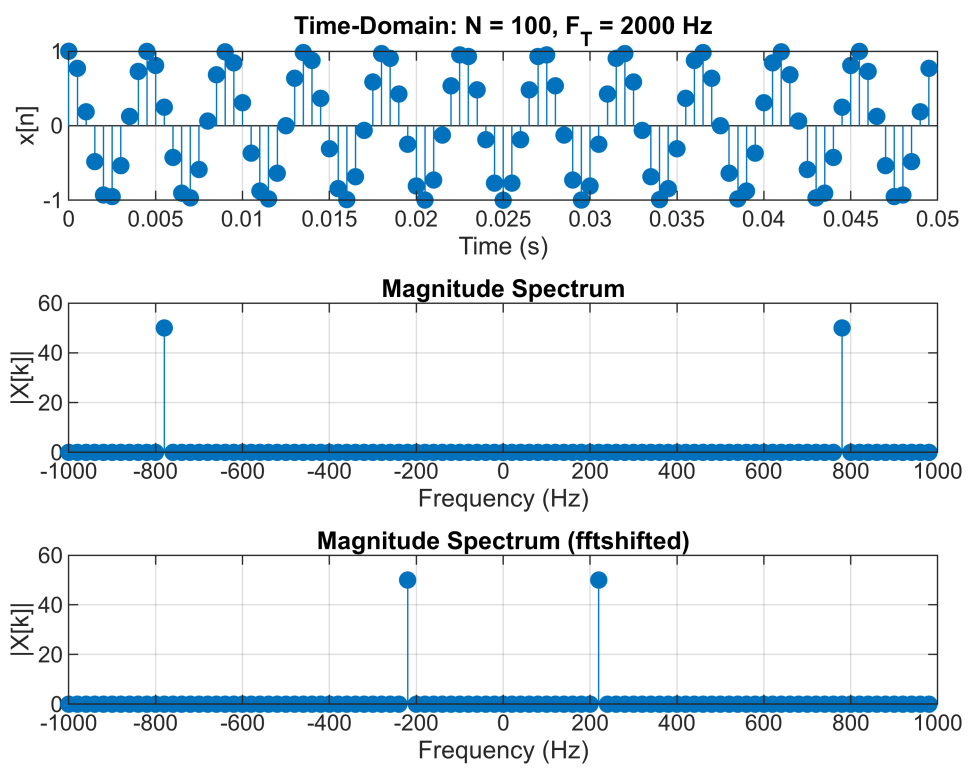
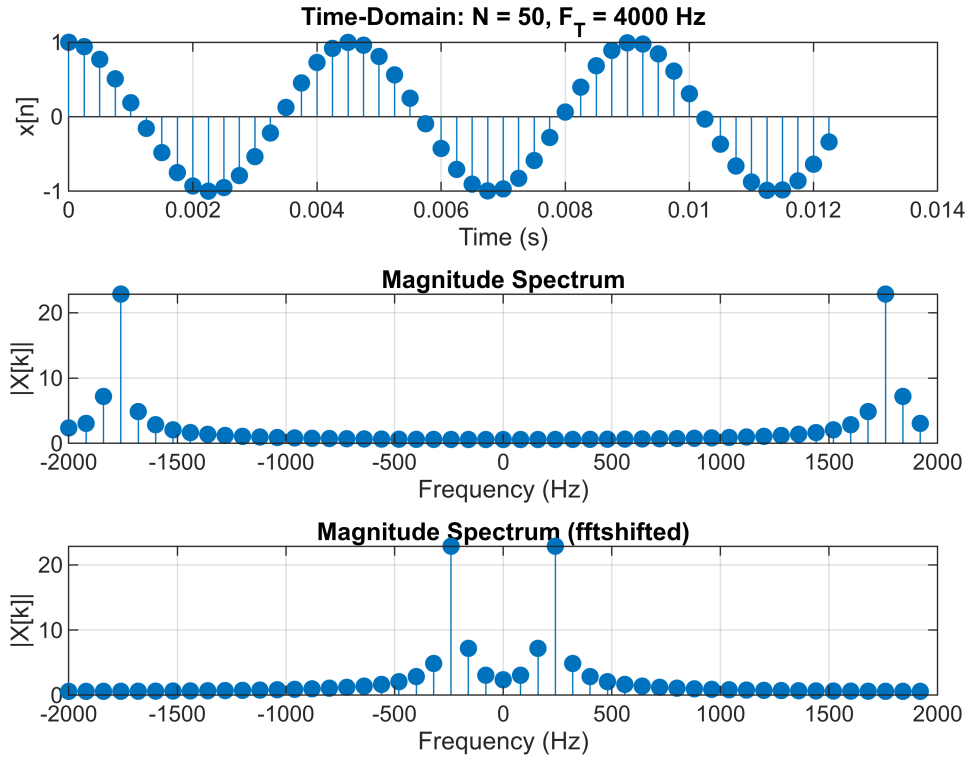
3. (10 points) The DFT is taken of N samples of $x(t) = \cos(2\pi(220)t)$ acquired with a sampling rate of F_T. For each of the following pairs of N and F_T : Use MATLAB to compute and stem plot the sample sequence with the horizontal axis labeled in units of time, not the sample index. Use MATLAB to compute and stem plot the magnitude of the DFT. Plot the DFT magnitude after using fftshift, and label the horizontal axis in units of Hz.

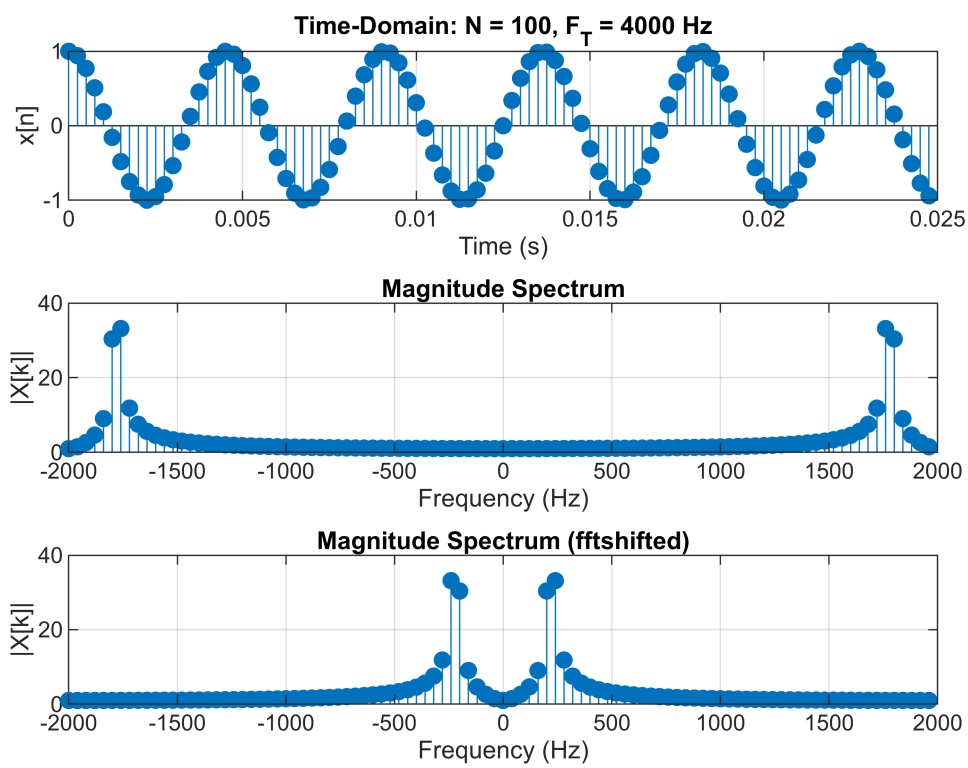
a.) N = 50 and $F_T = 2000$
b.) N = 100 and $F_T = 2000$
c.) N = 50 and $F_T = 4000$
d.) N = 100 and $F_T = 4000$
e.) N = 50 and $F_T = 1000$
f.) N = 100 and $F_T = 1000$

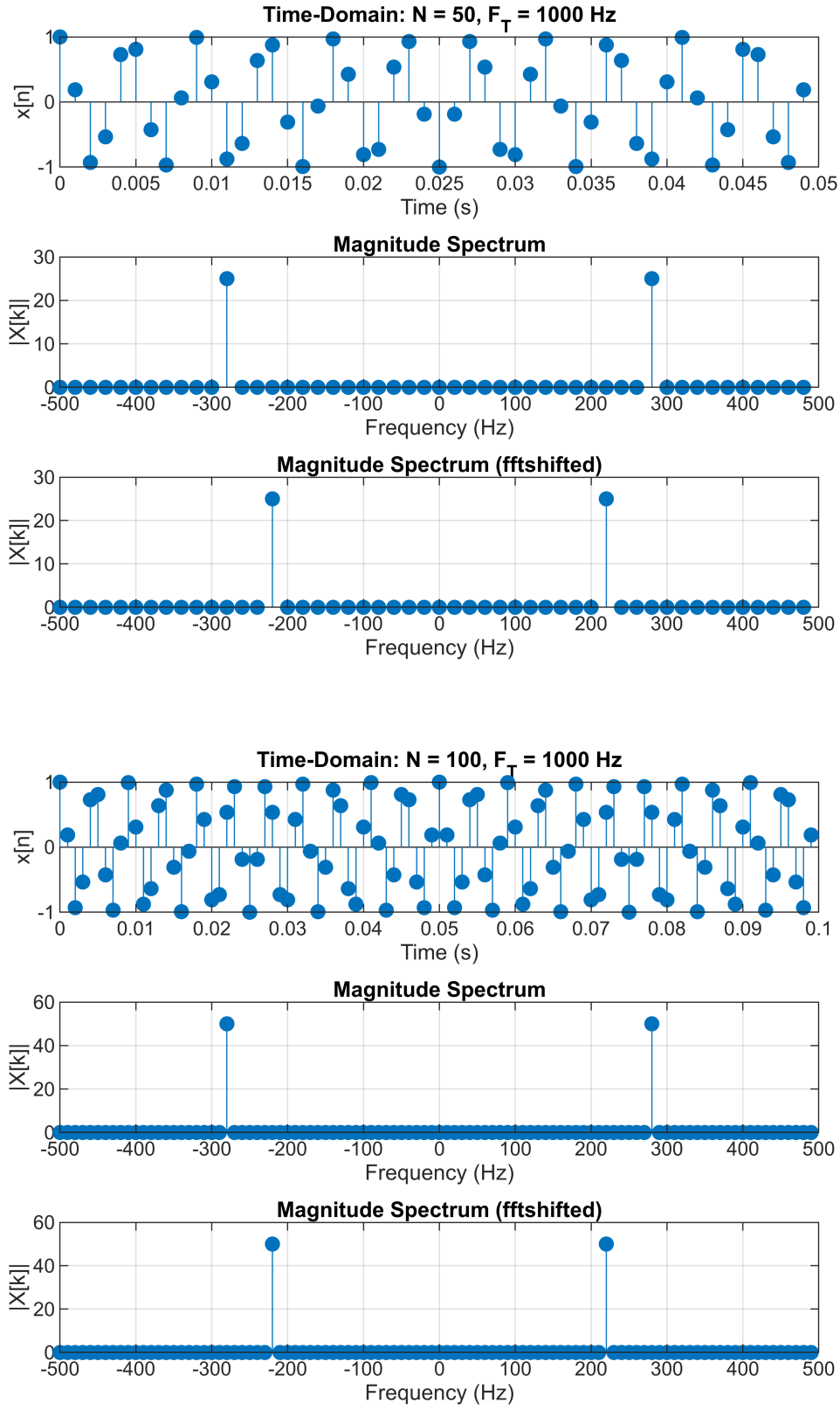
```
FT= [2000,2000,4000,4000,1000,1000];
N= [50,100,50,100,50,100];
for i=1:6
    T = 1 / FT(i);
    t = (0:N(i)-1) * T;
    x = cos(2*pi*220 * t);

    X = fft(x);
    Xshift = fftshift(X);
    f = (-N(i)/2:N(i)/2-1) * (FT(i) / N(i));

    figure;
    subplot(3,1,1);
    stem(t, x, 'filled');
    xlabel('Time (s)');
    ylabel('x[n]');
    title(sprintf('Time-Domain: N = %d, F_T = %d Hz', N(i), FT(i)));
    grid on;

    subplot(3,1,2);
    stem(f, abs(X), 'filled');
    xlabel('Frequency (Hz)');
    ylabel('|X[k]|');
    title('Magnitude Spectrum');
    grid on;

    subplot(3,1,3);
    stem(f, abs(Xshift), 'filled');
    xlabel('Frequency (Hz)');
    ylabel('|X[k]|');
    title('Magnitude Spectrum (fftshifted)');
    grid on;
end
```


Time-Domain: N = 50, F_T = 2000 Hz
x[n]
Time (s)
Magnitude Spectrum
|X[k]|
Frequency (Hz)
Magnitude Spectrum (fftshifted)
|X[k]|
Frequency (Hz)


Time-Domain: N = 100, F_T = 2000 Hz
x[n]
Time (s)
Magnitude Spectrum
|X[k]|
Frequency (Hz)
Magnitude Spectrum (fftshifted)
|X[k]|
Frequency (Hz)


Time-Domain: N = 50, F_T = 4000 Hz
x[n]
Time (s)
Magnitude Spectrum
|X[k]|
Frequency (Hz)
Magnitude Spectrum (fftshifted)
|X[k]|
Frequency (Hz)


Time-Domain: N = 100, F_T = 4000 Hz
x[n]
Time (s)
Magnitude Spectrum
|X[k]|
Frequency (Hz)
Magnitude Spectrum (fftshifted)
|X[k]|
Frequency (Hz)


Time-Domain: N = 50, F_T = 1000 Hz
x[n]
Time (s)
Magnitude Spectrum
|X[k]|
Frequency (Hz)
Magnitude Spectrum (fftshifted)
|X[k]|
Frequency (Hz)
Time-Domain: N = 100, F_T = 1000 Hz
x[n]
Time (s)
Magnitude Spectrum
|X[k]|
Frequency (Hz)
Magnitude Spectrum (fftshifted)
|X[k]|
Frequency (Hz)